

Sorbonne Paris North University
Master in Financial Engineering and Modeling
Financial Econometrics

Econometric Analysis and Modeling of Lockheed Martin Stock

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Stock Overview

Company

Lockheed Martin (LMT) is a major player in the defense and aerospace industries.

Financial Relevance

- a stock exposed to economic and geopolitical cycles;
- volatility driven by contract announcements, military budgets, and international tensions;
- a relevant asset for risk and volatility analysis;
- a company included in the S&P 500 index, used as a proxy for the U.S. equity market.

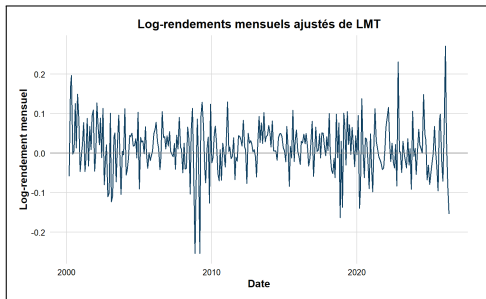


Preliminary Monthly Analysis

Objective

Before the daily analysis, monthly returns are examined to identify mean dynamics and a potential low-frequency ARCH effect.

LMT Monthly Returns



Statistic	Value	Interpretation
Mean	0.01269	positive
Min	-0.25424	extreme loss
Max	0.27099	extreme gain
Skewness	-0.1898	slight asymmetry
Kurtosis	5.1375	heavy tails
Jarque-Bera	61.858	non-normality
JB p-value	3.70×10^{-14}	normality rejected

Interpretation

Monthly returns remain non-normal, with heavy tails, but their dynamics are smoother than at the daily frequency.

Monthly Modeling: Summary Results

Stationarity

Test	Stat.	p-value	Conclusion
ADF	-7.3278	< 0.01	stationary
KPSS	0.1474	> 0.10	stationary

Selected Monthly ARIMA Model

ARIMA(2, 0, 2)

AIC	BIC	LB p-value	ARCH p-value
-833.709	-811.194	0.6712	0.0542

Monthly Forecast

The 12-month forecasts fluctuate around a small positive return, but the confidence intervals remain wide.

Static Monthly Risk

Level	VaR	Exceedances
95%	9.14%	16
99%	13.45%	6

Conclusion

The monthly analysis confirms stationarity and non-normality, but it remains too aggregated for a detailed analysis of volatility. The main analysis is therefore conducted at the daily frequency.

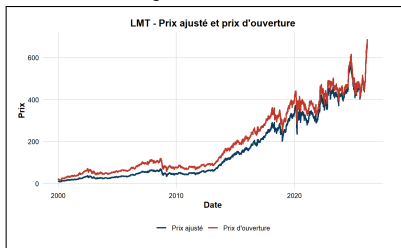
- ① Data processing and statistical analysis of returns
- ② Modeling the dynamics
 - ARIMA (conditional mean)
 - GARCH (conditional volatility)
- ③ Analysis of volatility dynamics
- ④ Risk measurement: VaR and Expected Shortfall
- ⑤ Model validation (backtesting)
- ⑥ Analysis of market exposure
 - Static CAPM
 - extension to a dynamic CAPM (rolling window)
- ⑦ Relationship with the VIX !
- ⑧ Conclusion and limitations

Data Processing and Statistical Analysis of Returns

Data

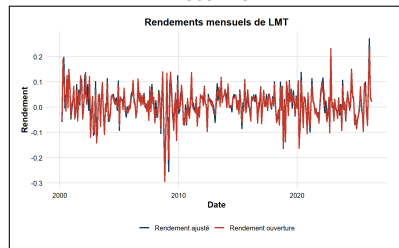
Daily series for **Lockheed Martin (LMT)** stock — Yahoo Finance, 2000–2026, **6 620 observations**

Adjusted Price



$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

Returns

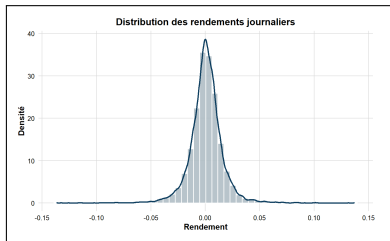


Test	Stat.	Parameters	p-value	Conclusion
ADF	-6.8863	lag = 6	< 0.01	Stationary
KPSS	0.0999	trunc.lag = 5	0.10	Stationary

Stationarity Tests — Monthly Returns

Return Properties

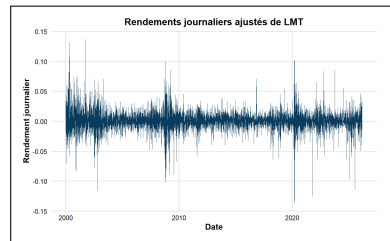
Return Distribution



	Mean	Min	Max	Skew	Kurt
Val	0.00059	-0.1365	0.1371	-0.173	12.12

Descriptive Statistics

Returns (Adjusted Close)



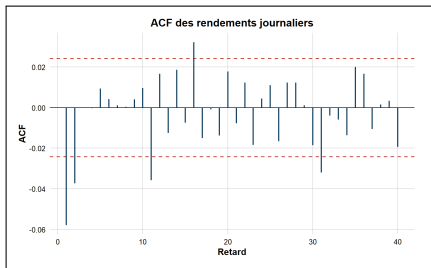
Test	Stat.	df	p-value	Conclusion
Jarque-Bera	22965	2	$< 10^{-16}$	Non-normality
Student t	3.036	—	< 0.01	Mean $\neq 0$

Statistical Tests

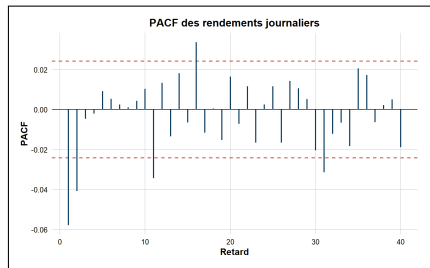
Non-normality, a statistically significant mean, and the presence of conditional heteroskedasticity.

ARIMA Model Identification

ACF of Returns



PACF of Returns



Comment

PACF and ACF: no significant structure beyond the second lag $\Rightarrow (p, q \leq 2)$

Selected ARIMA Model

Modeling the Mean Dynamics

the selected model is: **ARIMA(0,0,2)** $\Rightarrow X_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2}$

The dynamics are driven by recent shocks, indicating low predictability of the mean. The estimated coefficients are statistically significant, supporting the model specification.

AIC	BIC	LB	ARCH	JB
-36137.95	-36110.76	0.2036	$< 2.2 \times 10^{-16}$	0

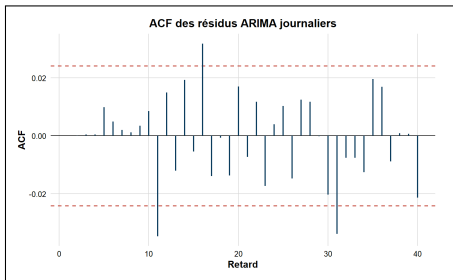
Parameter	Estimate	p-value
MA(1)	-0.0567	3.927×10^{-6}
MA(2)	-0.0356	0.0037
μ	0.00059	0.0008

Conclusion

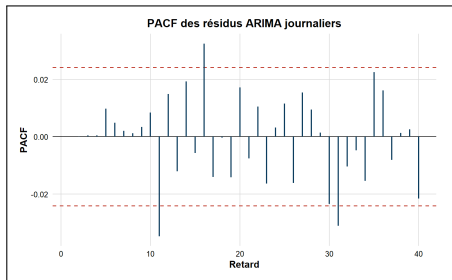
Uncorrelated residuals $\Rightarrow$ model valid for the conditional mean
ARCH effect present $\Rightarrow$ GARCH modeling is required

ARIMA Model Validation

ACF of Residuals



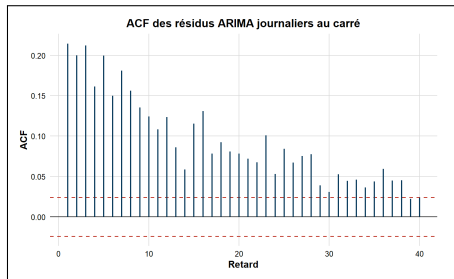
PACF of Residuals



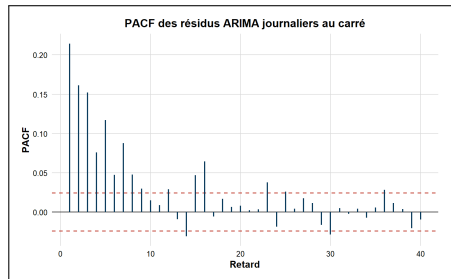
No significant autocorrelation $\Rightarrow$ residuals can be treated as white noise

Volatility Analysis (ARCH Effect)

ACF of Squared Residuals



PACF of Squared Residuals



Significant autocorrelations in squared residuals $\Rightarrow$ variance dependence $\Rightarrow$ presence of an ARCH effect.

Volatility Model Selection

Methodology

Several GARCH(1,1) specifications and asymmetric extensions are estimated under Gaussian and Student-t distributions.

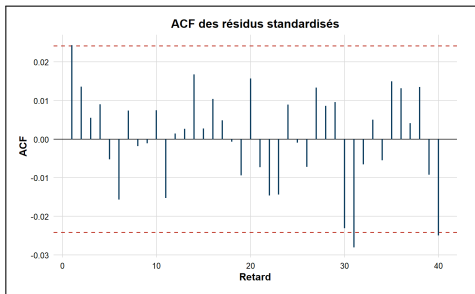
Model	AIC	BIC	LB	ARCH
TGARCH(1,1,1) - Student	3.345	3.350	0.810	0.978
GJR-GARCH(1,1,1) - Student	3.352	3.357	0.822	0.954
GARCH(1,1) - Student	3.353	3.357	0.797	0.975
TGARCH(1,1,1) - Gaussian	3.473	3.477	0.819	0.990
EGARCH(1,1,1) - Gaussian	3.473	3.477	0.802	0.990
GJR-GARCH(1,1,1) - Gaussian	3.482	3.487	0.820	0.989
GARCH(1,1) - Gaussian	3.484	3.487	0.806	0.991
EGARCH(1,1,1) - Student	9.994	9.999	0.0018	1.000

Conclusion

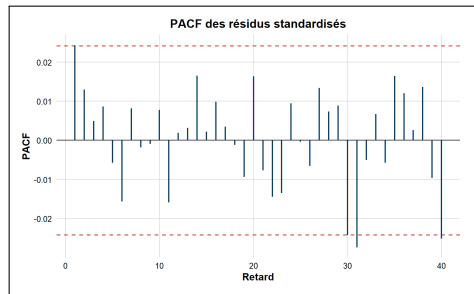
The **TGARCH(1,1,1) model with Student-t innovations** is selected because it minimizes the information criteria (AIC/BIC).

Graphical Model Validation

ACF of Standardized Residuals



PACF of Standardized Residuals



No significant structure $\Rightarrow$ the residuals behave like white noise.

Parameters and Properties of the TGARCH(1,1) Student-t Model

Model Parameters

Parameter	Value	Description
ω	0.0237	Baseline volatility
α	0.0836	Reaction to recent shocks
β	0.9220	Persistence
η	0.2522	Asymmetry
ν	4.89	Student-t distribution

TGARCH Model Rationale

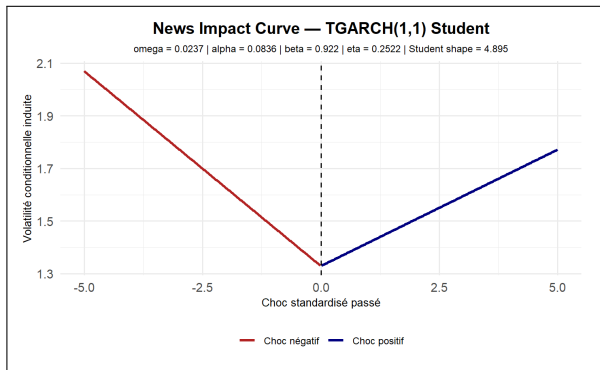
- **Conditional and persistent** volatility
- Asymmetry ($\eta > 0$) $\Rightarrow$ negative shocks amplify volatility
- Student-t distribution $\Rightarrow$ accounts for **heavy tails**

Process Properties

- Persistence : $0.9833 < 1 \Rightarrow$ highly persistent but **stationary** volatility
- Half-life: ≈ 41.1 days $\Rightarrow$ slow dissipation of shocks
- Long-run volatility: $\approx 1.42\%$ $\Rightarrow$ consistent with a large-cap stock

News Impact Curve — Volatility Asymmetry

News Impact Curve — TGARCH(1,1) Student



Interpretation

The curve measures the impact of a past shock on future volatility. Its **asymmetry** is the key result.

Main Result

Shock	Magnitude	Implied vol.
Negative	-5σ	$\approx 3.2\%$
Neutral	0	$\approx 1.4\%$
Positive	$+5\sigma$	$\approx 2.0\%$

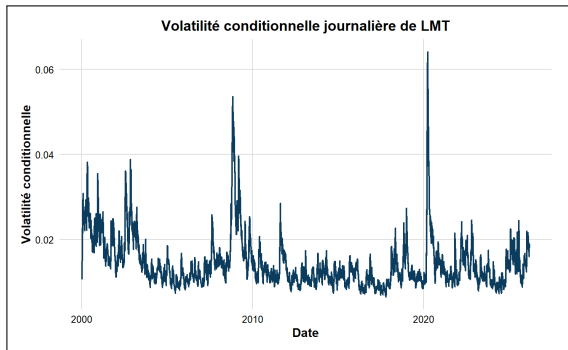
A negative shock generates **60 % more volatility** than a positive shock of the same magnitude.

Why?

The effective coefficient rises from $\alpha = 0.084$ for a positive shock to $\alpha + \eta = 0.336$ for a negative shock — making the model **4× more responsive** to bad news.

Event-Based Analysis of Conditional Volatility

Annotated GARCH Volatility (2000–2025)



Identified Events

Date	Event
2000	Dot-com bubble burst
Sep. 2001	September 11 attacks
Mar. 2003	Invasion of Iraq
Oct. 2008	Global financial crisis
Aug. 2011	U.S. debt crisis / Sequestration
Mar. 2020	COVID-19 pandemic
Mar. 2022	Russian invasion of Ukraine
Oct. 2023	Hamas–Israel conflict
Oct. 2024	F-35 shock / LMT Q3 results
2025	Trump/DOGE budget uncertainty

Key Observation

LMT volatility reacts to two types of shocks :

- **Geopolitical shocks:** increase driven by expectations of new contracts
- **Systemic shocks:** increase driven by budget uncertainty

Economic Interpretation of Volatility Spikes

LMT Exposure to Event Risk

As the world's largest defense contractor (~75 % of revenue from the U.S. Department of Defense), LMT exhibits asymmetric sensitivity to macroeconomic and geopolitical events.

Event	Date	Peak	Mechanism
Dot-com bubble	2000	≈ 3.8%	Broad market sell-off; defensive fundamentals remained intact
9/11 attacks	Sep. 2001	≈ 4.5%	Defense budgets up +10% as early as FY2002
Iraq invasion	Mar. 2003	≈ 4.5%	First F-35 orders; uncertain conflict duration
Lehman crisis	Oct. 2008	≈ 5.0%	LMT down -51.4% in 7 months; Obama-era cuts
Sequestration	Aug. 2011	≈ 3.0%	Budget Control Act: automatic DoD spending cuts

Event	Date	Peak	Mechanism
COVID-19	Mar. 2020	≈ 5.9%	LMT down -37.1% in 40 days; supply-chain disruption
Ukraine invasion	Mar. 2022	≈ 6.7%	Record peak: European rearmament, +40%
Hamas-Israel	Oct. 2023	≈ 2.8%	Multiple conflicts; renewed Patriot orders
F-35 shock (Q3)	Oct. 2024	≈ 4.2%	Losses of \$1.6bn; negative cash flow
Budget Trump	2025	≈ 2.6%	DOGE ambiguity versus a \$1.5tn budget

The TGARCH effect ($\eta = 0.2522$) confirms that **negative** shocks amplify volatility more strongly — consistent with LMT's defensive profile.

Conditional Volatility Forecast — TGARCH(1,1,1)

Estimated and Forecast Volatility (Sep. 2025 – 2026)

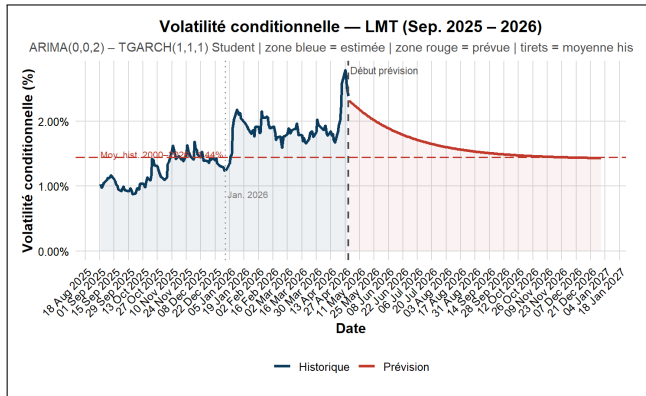


Chart Interpretation

- **Blue area** : estimated volatility (Sep.–Dec. 2025) period of particular interest !!
- **Red area** : forecast volatility (2026)
- Gradual return toward the long-run level ($\approx 1.42\%$)
- High persistence (≈ 0.98) slows the dissipation of shocks

Conclusion

LMT volatility in 2026 is expected to **converge slowly** toward its long-run mean, without an abrupt reversal, in line with the estimated half-life of ≈ 41 days.

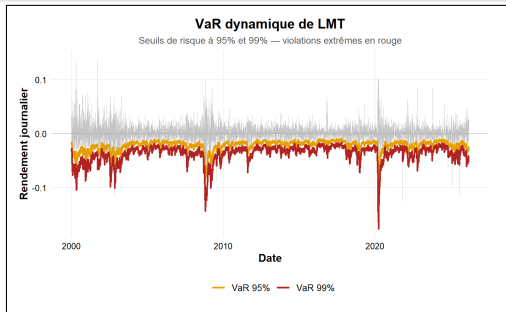
Dynamic Value-at-Risk

Principle

VaR estimates the maximum expected loss at a given confidence level, dynamically adjusting through the ARIMA mean and TGARCH volatility.

$$VaR_{\alpha,t} = -(\mu_t + \sigma_t q_{\alpha})$$

μ_t ARIMA conditional mean
 σ_t TGARCH conditional volatility
 q_{α} standardized Student-t quantile

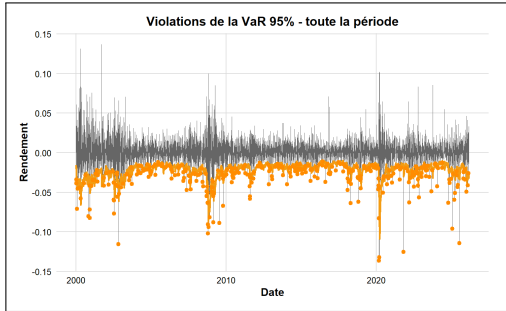


Interpretation

The 95% VaR sets a threshold exceeded in approximately 5% of cases, whereas the 99% VaR is more stringent.

VaR Validation

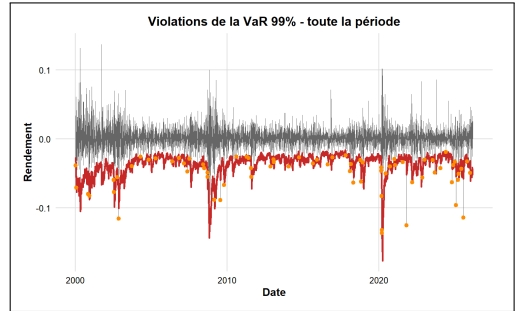
VaR 95%



Exceptions	Observed freq.	Theoretical freq.
348	5.26%	5.00%

Kupiec : $p = 0.342$ Christoffersen : $p = 0.115$

VaR 99%



Exceptions	Observed freq.	Theoretical freq.
73	1.10%	1.00%

Kupiec : $p = 0.409$ Christoffersen : $p = 0.202$

The Kupiec and Christoffersen tests do not reject the VaR model: both the frequency and independence of violations are consistent with the model.

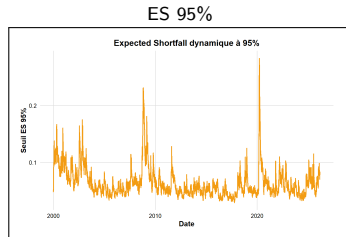
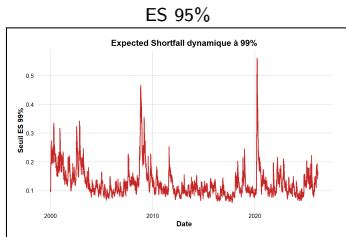
Dynamic Expected Shortfall

Principle

Expected Shortfall measures the average conditional loss beyond the VaR threshold. It is computed dynamically from the ARIMA-TGARCH model with Student-t innovations.

$$ES_{\alpha,t} = - \left(\mu_t + \sigma_t \cdot \frac{f(q_{\alpha})}{\alpha} \cdot \frac{\nu + q_{\alpha}^2}{\nu - 1} \right)$$

μ_t, σ_t conditional mean and volatility
 q_{α}, ν Student-t quantile and degrees of freedom



The McNeil–Frey test validates Expected Shortfall at the 95% and 99% levels (bootstrap p-values: 0.269 and 0.349).

CAPM — Modeling Market Exposure

Model

The CAPM can be estimated empirically using a regression that relates an asset's excess return to the market's excess return.

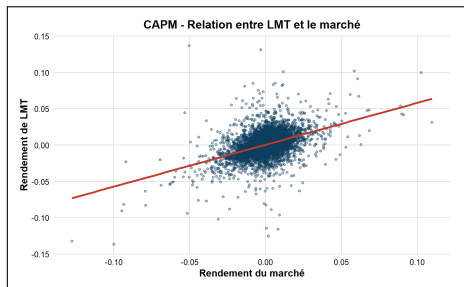
$$R_{LMT,t} - R_{f,t} = \alpha + \beta(R_{m,t} - R_{f,t}) + \varepsilon_t$$

Empirical Results

- $\alpha = 0.00042$ (significant)
- $\beta = 0.575$ (highly significant)
- $R^2 = 19.6\%$

Interpretation

- $\beta < 1 \Rightarrow$ LMT is **defensive**
- Low $R^2 \Rightarrow$ a large share of idiosyncratic risk
- $\alpha > 0 \Rightarrow$ slight outperformance



Dynamic CAPM

Principle

The classical CAPM assumes a constant beta. In practice, market exposure changes over time.

Modeling Approach

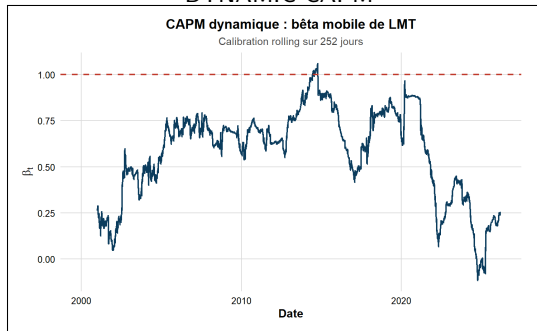
$$R_{LMT,t} - R_f = \alpha_t + \beta_t(R_{m,t} - R_f) + \varepsilon_t$$

Beta is estimated using a rolling window.

Interpretation

- $\beta_t > 1$: high market exposure
- $\beta_t < 1$: defensive stock
- Variations: changes in the risk regime

DYNAMIC CAPM



Beta varies over time, reflecting changes in market exposure.

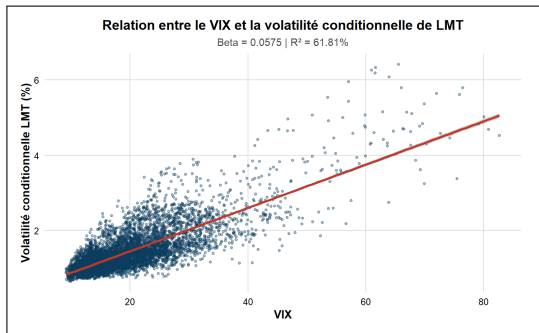
LMT Volatility and the VIX

Regression of Conditional Volatility on the VIX

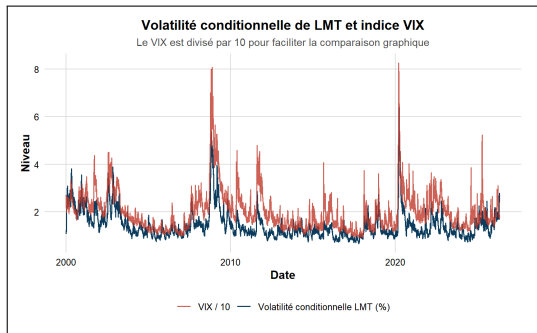
$$\sigma_t = \alpha + \beta VIX_t$$

$\beta = 0.0575$ | $R^2 = 61.8\%$ | strong positive relationship

Regression: Conditional Volatility vs. VIX



Time-Series Comparison: LMT Volatility and VIX



The VIX explains a substantial share of LMT's conditional volatility: volatility rises with market stress.

Conclusion

Main Contributions

- A series of **6 620 daily observations** (2000–2026), covering five major geopolitical cycles
- **ARIMA(0,0,2)** model: mean dynamics driven by recent shocks, with low predictability
- **TGARCH(1,1,1) with Student-t innovations**: best AIC/BIC trade-off among eight tested specifications
- **Confirmed asymmetry** ($\eta = 0.2522$): negative shocks amplify volatility more strongly
- **High persistence** (0.9833), with a half-life of ≈ 41 days: consistent with defense-sector cycles
- **Dynamic VaR and ES** validated by the Kupiec, Christoffersen, and Acerbi–Szekely tests
- **CAPM**: $\beta = 0.575$ confirms the defensive profile; dynamic β_t reveals changing regimes

Key Message

LMT is a **defensive yet event-sensitive** stock: its volatility is structurally low and persistent, and reacts asymmetrically to geopolitical and budgetary shocks — dynamics that the ARIMA-TGARCH model captures effectively.

Limitations and Further Research

Limitations

- Symmetric distribution (does not capture skewness)
- Parametric VaR is sensitive to the chosen distribution
- Linear CAPM (instability of β_t)
- Unobserved factors (geopolitics, announcements)

Further Research

- **Regime-Switching Models (MS-GARCH)**
Distinguish between market phases (calm versus crisis) with different volatility dynamics
- **Skewed Student-t Distribution**
Incorporate return asymmetry to improve risk modeling
- **Exogenous Variables (VIX, GPR)**
Add market-stress and geopolitical-risk indicators to the volatility dynamics
- **Hybrid Models**
Capture nonlinear dynamics not explained by GARCH models

- Kupiec, P. (1995). *Techniques for verifying the accuracy of risk measurement models*. Journal of Derivatives, 3(2), 73–84.
- Christoffersen, P. (1998). *Evaluating interval forecasts*. International Economic Review, 39(4), 841–862.
- McNeil, A. J., & Frey, R. (2000). *Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach*. Journal of Empirical Finance, 7(3–4), 271–300.
- Fama, E. (1970). *Efficient Capital Markets: A Review*. Journal of Finance, 25(2), 383–417.

Questions?

Thank you.